

JASWANTH KADALI

Mechanical Engineering • Robotics & Autonomous Systems • Drone Engineering
Bhimavaram, India • +91 83417 39119 • jaswanthkadali0362@gmail.com • LinkedIn • Portfolio

OBJECTIVE

Final-year Mechanical Engineering student specializing in robotics, autonomous drones, and vehicle systems, with hands-on experience in CAD design, FEA simulation, embedded electronics, and field validation under real-world operating conditions. Skilled at delivering complete systems from concept through fabrication and testing, including an autonomous precision-agriculture drone and award-winning off-road vehicles. Seeking a full-time role or internship in field-deployed robotic platforms.

EDUCATION

SRKR Engineering College — Bhimavaram, Andhra Pradesh

B.Tech in Mechanical Engineering • Final Year • CGPA: 7.5 / 10.0

TECHNICAL SKILLS

- **Mechanical CAD & Simulation:** Siemens NX, SOLIDWORKS, Fusion 360, Solid Edge, AutoCAD, Siemens MCD (digital twin), ANSYS FEA, MATLAB
- **Design Standards & Manufacturability:** GD&T (ASME Y14.5), DFM/DFA (Design for Manufacturing & Assembly), Parametric Modeling & 2D Detailing (drafting, section views, sheet metal, surface modeling), Finite Element Analysis (structural & thermal), PLM/PDM Awareness (Teamcenter, Windchill)
- **Robotics & Autonomous Systems:** ROS 2 (Jazzy), Gazebo simulation, Micro-ROS, URDF/Xacro, Navigation2, sensor fusion
- **Drones & UAV:** Pixhawk / ArduPilot, frame design, PID flight tuning, GPS waypoint navigation
- **Embedded Systems:** ESP32, Arduino UNO, C/C++, PWM control, motor drivers, sensor integration
- **Vehicle & Mechanical Systems:** Chassis design, Ackermann geometry, structural durability
- **Manufacturing & Prototyping:** FDM 3D printing (PLA/PETG/ABS/TPU), CNC-ready part design, mechanical rigs/mounts/brackets, Cura, PrusaSlicer
- **Computer Vision & Machine Learning:** Deep learning and machine learning algorithms, object detection, image classification, computer vision pipelines
- **Quality Tools & Methodologies:** 7 QC Tools, Root Cause Analysis (RCA), 5S, 8D Reporting
- **Lean Manufacturing:** Kaizen, line balancing, cycle time reduction, waste elimination
- **Manufacturing Processes:** CNC machining fundamentals, sheet metal forming, injection molding, assembly line balancing
- **Quality Standards:** ISO 9001, awareness of aerospace/defense standards (AS9100)
- **Process Documentation:** Standard Operating Procedures (SOP), Work Instructions (WI), Process Flow Charts (PFC)

KEY PROJECTS

NIDAR — Precision Agriculture Drone System | *National Innovation Challenge for Drone Application & Research, Greater Noida*

- Designed and built two purpose-built quadcopters for precision agriculture: a scouting drone to identify infected or stressed plants, and a dedicated spraying drone to treat them.
- Scouting drone: integrated an NVIDIA Jetson Nano running ROS 2 for real-time onboard computer vision, flagging crop stress and disease early for targeted intervention.
- Spraying drone: built a full-payload sprayer system with GPS waypoint navigation to autonomously treat the areas identified by the scouting drone.
- Both drones share a Pixhawk Cube Orange flight controller (triple-redundant IMU, ArduPilot autonomy) and a custom impact-absorbing frame; validated PID flight tuning on a dedicated test bench.

Off-Road ATV — Design & Manufacturing | *Team Godavari, FOMBI National Off-Road ATV Championship*

- Led chassis design and manufacturing of a student off-road ATV, focused on structural durability and real-world manufacturability.
- Worked extensively on assembly and testing for sustained rough-terrain operation.
- Result: Winner, FOMBI 2K25; Runner-up, FOMBI 2K24.

RC 4WD Vehicle with Ackermann Steering

- Designed and assembled a 4WD RC vehicle on Arduino UNO, implementing Ackermann steering geometry to improve turning stability and control.
- Gained hands-on experience with motor calibration, steering linkages, and mechanical alignment.

Impact-Absorbing Drone Arm Design

- Designed and optimized drone arm structures for improved impact absorption, focusing on structural strength and load distribution.
- Iterated designs through CAD-based analysis followed by physical validation testing.

ESP32-Based Vibration & Thermal Monitoring System

- Built a real-time monitoring system using ESP32 with vibration and thermal sensors to detect abnormal ESC conditions.
- Enabled early fault detection for robotic and electromechanical systems operating under stress.

LPG Gas Leakage Detection System

- Developed a household/industrial safety system using MQ-05 sensors for gas leak detection.
- Implemented sensor integration and system logic for reliable continuous operation.

ACHIEVEMENTS & COMPETITIONS

- Winner — FOMBI 2K25, National Off-Road ATV Championship (ISNEE Motorsports), Team Godavari
- Runner-up — FOMBI 2K24, National Off-Road ATV Championship (ISNEE Motorsports), Team Godavari — Sagar Institute of Science, Technology & Research, Bhopal (27 Sep–1 Oct 2024)
- Runner-up — SRKR Go-Kart Championship, Team Markandeya, SRKR Engineering College (23–26 Jan 2025)
- Participant — AGKC 2025 Season II, National-Level Go-Karting Championship, AITAM, Tekkali (4–7 Mar 2025)
- Participant — National Innovation Challenge for Drone Application & Research (NIDAR), Greater Noida

CERTIFICATIONS & WORKSHOPS

CAD & Simulation

- Siemens NX Mastery: Advanced Design & Applications — Siemens, via Coursera (Apr 2026)
- Intro to Siemens NX: Engineering Essentials & Part Design — Siemens, via Coursera (Nov 2025)
- SOLIDWORKS Advanced Sketching & Reference Geometry — Dassault Systèmes, via Coursera (Nov 2025)
- CAD Designing & ANSYS Analysis — Course Central

AI & Computer Vision

- Advanced Deep Learning Techniques for Computer Vision — MathWorks, via Coursera (Mar 2026)
- Introduction to Deep Learning for Computer Vision — MathWorks, via Coursera (Feb 2026)

Robotics, Drones & Embedded Systems

- Autonomous Mobile Robotics (AMR) Mastery Program — MechaMind Labs, 45-day hands-on program on robot design, simulation & autonomous systems with ROS 2 (Jan–Mar 2026)
- MicroROS and ESP32 WiFi-Controlled Robot Hands-On using ROS2 — Udemy (Nov 2025)
- Hands-on Design, Integration & Assembly of Drones — Binford Research Labs Pvt. Ltd.
- Advanced Drone Technology (Air Taxi) Workshop — I-STEM CeNSE, Bharat Space Education Research Centre & IISc Bengaluru (Jul 2025)
- ArduBotics Robotics — 2-Day National Workshop, Technex'25, IIT (BHU) Varanasi, delivered by Innovians Technologies (Feb 2025)

Quality & Process

- Six Sigma Yellow Belt — LinkedIn Learning (Jul 2026)
- Reverse Engineering Foundations: Product Design — LinkedIn Learning (Jul 2026)

Mentorship

- ATL Mentorship Program (3 Days) — JNTUK, Kakinada